

Poojitha Ratna Raghavarapu

PH: +13415002624 | EMAIL: raghavarapupoojitha@gmail.com | [LinkedIn](#)

SUMMARY

Software Engineer with over two years of experience in developing AI-powered and enterprise applications using Python and Java for leading organizations including Cardinal Health and Tata Consultancy Services. Achievements include building Generative AI and RAG-based solutions that improved knowledge retrieval efficiency by 75% and developing scalable RESTful services that enhanced application performance and operational productivity

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, Scala, SQL, C, C++, C#, .Net, GoLang, Rust
Artificial Intelligence & GenAI: LLMs, RAG, LangChain, LangGraph, OpenAI, Azure OpenAI, Prompt Engineering, AI Agents, Semantic Search, Embeddings, Vector Search, NLP
Backend Technologies: FastAPI, Flask, REST APIs, Microservices, JSON, Postman
Cloud Platforms: Microsoft Azure (Azure OpenAI Service, Azure AI Search, Azure Blob Storage, Azure App Services, Azure Key Vault), AWS (S3, EC2, Cloud Formation, Cloud Watch, Data pipeline, Glue, Athena, DynamoDB, Kinesis), GCP
Web Technologies: HTML, CSS3, React, React Native, AngularJS, Nodejs, Vue.js, Next.js, Flask, Redux
Databases & Vector Stores: SQL Server, MySQL, PostgreSQL, Azure AI Search, Pinecone, ChromaDB, FAISS
Java Technologies: Core Java, Spring Boot, Spring MVC, Hibernate, JDBC, Maven, JUnit
Data Processing: Pandas, NumPy, ETL Pipelines, Data Transformation, Data Cleaning
CI/CD & DevOps: Git, GitHub, GitHub Actions, Docker, Jenkins
Methodologies: Agile, Scrum, SDLC, Test-Driven Development (TDD), Code Reviews

WORK EXPERIENCE

AI Engineer Cardinal Health - Engineered an enterprise Healthcare Knowledge Assistant using Azure OpenAI, LangChain, and FastAPI, enabling intelligent access to 50K+ healthcare documents and reducing information retrieval time by 75% - Built end-to-end RAG pipelines with Azure AI Search and vector embeddings, improving answer relevance by 40% and retrieval accuracy by 35% - Designed automated document ingestion and indexing frameworks for PDFs and compliance records, processing 50K+ documents and reducing manual lookup effort by 60% - Developed RESTful APIs with FastAPI and optimized AI workflows, reducing response latency by 30% and document onboarding time by 50% - Implemented prompt engineering and GPT-4 guardrails, lowering hallucinated responses by 25% and enhancing reliability of AI-generated outputs - Containerized applications with Docker and supported Azure cloud deployments, improving deployment efficiency by 40% while maintaining 99.5% service availability	06/2025-Present
Software Engineer Tata Consultancy Services - Developed and maintained Java-based enterprise applications using Core Java, Spring Framework, and OOP principles, improving application performance by 25% - Designed and optimized SQL queries, stored procedures, and RESTful APIs, reducing database response times by 30% and improving data exchange efficiency by 35% - Performed code reviews, unit testing, and defect resolution, reducing production issues by 20% and enhancing application reliability - Utilized Git, CI/CD pipelines, and AWS cloud services to support application deployments, monitor production systems, and resolve 95% of critical incidents within SLA targets	06/2022-08/2023

ACADEMIC TECHNICAL PROJECT

Animal Image Classification using Deep Learning and Transfer Learning - Developed a deep learning-based animal image classification system using Python, TensorFlow, and transfer learning models (VGG16, ResNet50, EfficientNetB7) to classify 10 categories from 26K+ images, achieving 97.16% test accuracy - Built an end-to-end machine learning pipeline for data preprocessing, augmentation, training, and model evaluation using TensorFlow/Keras, NumPy, Pandas, and Matplotlib, improving classification reliability and automating wildlife image analysis	01/2025
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EDUCATION AND COURSEWORK

Master of Computer Science Governors State University – Chicago IL	08/2023-05/2025
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CERTIFICATIONS

Microsoft Azure AI Fundamentals	07/2025
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